1. It
appears that the size of the batch affects the scrap percentage.

Which
variable is likely to be the CTQ and which variable the influence factor?

1 point

- Both batch size and scrap percentage are influence factors.
- Batch size is the CTQ and scrap percentage the influence factor.
- Batch size is the influence factor and scrap percentage the CTQ.
- Both batch size and scrap percentage are CTQs.

2. You
want to know whether the Throughput Time is influenced by the day of the week
that the request is submitted.

Which
test of the following four options is appropriate for this purpose?

1 point

- Logistic regression.
- ANOVA.
- Regression.
- Chi-square.

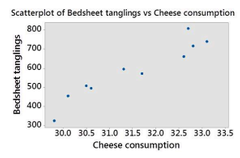
3. You
obtain a p-value equal to 0.01.

Using
a significance level of 0.05, what is the correct conclusion?

1 point

- Do not reject the null hypothesis.
- Reject the alternative hypothesis.
- Do not reject the alternative hypothesis.
- Reject the null hypothesis.

1 point

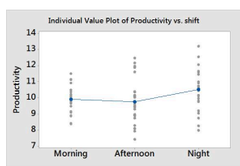


4. This scatterplot
implies that cheese consumption per capita (Cheese consumption) is
related with the number of people who died by becoming tangled in their
bedsheets (Bedsheet tangling).

What
is a plausible explanation for this correlation?

- Over time both variables have moved in the same direction.
- Eating cheese is dangerous.
- Due to a causal relationship between these variables, this pattern appears.
- Many people who die from tangling's have first eaten a lot of cheese.

1 point



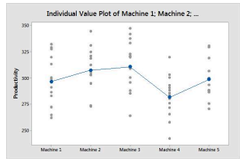

```
One-way ANOVA: Productivity vs Shift
Null hypothesis         All means are equal
Alternative hypothesis  At least one mean is different
Significance level      α = 0.05
Equal variances were assumed for the analysis.

Analysis of Variance
Source  DF   Adj SS  Adj MS  F-Value  P-Value
Factor   2    8.152   4.076     2.87    0.063
Error   72  102.362   1.422
Total   74  110.515

Model Summary
      S   R-sq  R-sq(adj)  R-sq(pred)
1.19235  7.38%      4.80%       0.00%

Means
Factor                   N    Mean  StDev       95% CI
Productivity_morning    25   9.902  0.815  ( 9.426; 10.377)
Productivity_afternoon  25   9.755  1.403  ( 9.280; 10.231)
Productivity_night      25  10.516  1.278  (10.041; 10.992)
Pooled StDev = 1.19235
```

5. In
the figure we see the productivity for each shift (morning, afternoon, night).
ANOVA is used to analyze the
difference in productivity between the shifts.

Based on the outputs presented, what is the correct conclusion?

- The low *R*-squared suggests that we should have performed another test to compare shifts.
- Although we observe small differences between shifts, the evidence in the data is not significant, indicated by the *p*-value that is larger than 0.05.
- The
  mean productivity
  differs significantly
  between shifts.
- Although we observe small differences between shifts, the evidence in the data is not significant, indicated by the *R*-square that is larger than 5%.

1 point




```
One-way ANOVA: Productivity vs. Machine
Null hypothesis         All means are equal
Alternative hypothesis  At least one mean is different
Significance level      α = 0.05

Analysis of Variance
Source  DF  Adj SS  Adj MS  F-Value  P-Value
Factor   4    7615  1903.7     4.37    0.003
Error   70   30463   435.2
Total   74   38078

Model Summary
      S    R-sq  R-sq(adj)  R-sq(pred)
20.8613  20.00%     15.43%       8.16%

Means
Factor      N    Mean  StDev       95% CI
Machine 1  15  296.32  23.46  (285.58; 307.06)
Machine 2  15  307.11  19.20  (296.37; 317.86)
Machine 3  15  310.37  24.01  (299.62; 321.11)
Machine 4  15  281.52  19.35  (270.78; 292.26)
Machine 5  15  298.51  17.49  (287.77; 309.25)

Pooled StDev = 20.8613
```

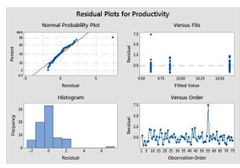
6. In the figure we see the productivity for each machine (1 through 5). ANOVA is used to analyze the effect of machines on the productivity

Based on the outputs presented, what is the correct conclusion?

1 point

- The mean productivity differs significantly between machines. However, besides the machine, other influence factors are also important as indicated by the R-square.
- Although we observe differences between machines, the evidence in the data is not significant, indicated by the small *p*-value.
- The mean productivity differs significantly between machines. Furthermore, machine is a very important influence factor as indicated by the high R-square.
- Although we observe differences between machines, the evidence in the data is not significant, indicated the low *R*-square.

1 point



7. The
output is a residual analysis corresponding to an ANOVA, which studies the
effect of shift (morning, afternoon, night) on
the
CTQ: Productivity. What is a valid conclusion based on the shown output?

- The residuals show an outlier, therefore the results of the ANOVA (p-value, R-squared) are not reliable.
- The residuals are clearly nonnormal, so the ANOVA is valid.
- The results of the ANOVA (p-value, R-squared) are reliable.
- The influence factor is not categorical, therefore ANOVA is not the appropriate test.

1 point

```
Kruskal-Wallis Test: Remarks versus Office
Kruskal-Wallis Test on Remarks

Office     N  Median  Ave Rank      Z
North     25  10.611      51.2   3.72
East      25   9.930      36.0  -0.55
West      25   9.246      26.7  -3.17
Overall   75              38.0

H = 16.13  DF = 2  P = 0.000
```

8. CTQ: Number of remarks per report

Influence factor: the office
handling the report (North, East, West)

We have performed a Kruskal-Wallis test to see if the number
of remarks differs across offices.

What
can we conclude?

- The number of remarks has a significant effect on office.
- There is insufficient evidence that the medians of the groups are different.
- There is no significant relationship between Office and Remarks.
- There are differences in the number of remarks between the offices.

1 point

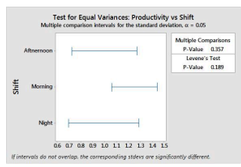
```
Two-sample T for Weekday vs Weekend-day

             N   Mean  StDev  SE Mean
Weekday     25  20.92   5.35      1.1
Weekend-day 25   2.96   1.17     0.23


Difference = μ (Weekday) - μ (Weekend-day)
Estimate for difference:  17.96
95% CI for difference:  (15.71; 20.21)
T-Test of difference = 0 (vs ≠): T-Value = 16.39
P-Value = 0.000  DF = 26
```

9. We have performed a 2-sample t-test
on the amount of e-mails somebody receives on week-days and weekend-days.

Given the output, which statement
is true?

- An ANOVA test should have been performed.
- There is no evidence for the difference in means being more than 0 (zero).
- The standard deviations differ significantly.
- The means differ significantly.

1 point



10. CTQ: Productivity.

Influence factor: Shift (morning, afternoon, night).

You performed an equal variances
test on the data
from which the output is given.

In the next step you want to study
the difference in average productivity between the three shifts. To this end, which
analysis should follow?

- Perform another test for equal variances not assuming equal variance.
- Perform an ANOVA assuming equal variances.
- Perform another test for equal variances assuming equal variance.
- Perform an ANOVA not assuming equal variances.